

COMPUTER ENGINEER • FULL-STACK DEVELOPER

Ms. Natthamon Kosem

Full-Stack & Backend Software Engineer | B.Eng Computer | 4+ Years Exp | Bangkok, TH

Phone: 099-564-9387
Email: nutkosem@gmail.com
Location: Nonthaburi / Bangkok, Thailand
GitHub: natthamon0709/portfolio

PERSONAL PROFILE

Computer engineer and full-stack software engineer with **4+ years of professional experience** across the entire product lifecycle — from reactive web frontends (Vue/Nuxt, React, TypeScript) to mission-critical backend microservices, RESTful APIs, and enterprise relational databases (**MySQL & Oracle DB**). Experienced across all testing levels (unit, functional, integration, system, and user acceptance) with automated CI/CD and industrial IoT integration.

AREAS OF EXPERTISE

CORE LANGUAGES

Python | JavaScript (ES6+) | TypeScript | PHP | SQL | C / C++ | HTML5 / CSS3

BACKEND & APIS

RESTful API Architecture | Node.js (Express) | PHP (Laravel MVC) | Microservices | OpenAPI / Swagger Specs | Message Queues & Workers

DATABASES & PERSISTENCE

MySQL | Oracle Database | PL/SQL Stored Procedures | Schema Normalization | Query Optimization & Indexing | Transaction Integrity

FRONTEND & WEB

Vue.js / Nuxt 4 | React & Redux | Tailwind CSS | React Hook Form | jQuery / DOM | Responsive Design

DEVOPS, CLOUD & TESTING

GitLab CI/CD Pipelines | Git / GitHub | Docker Containerization | Postman Automated Testing | Unit & Functional Testing | Linux (Bash CLI) | Vercel Serverless

CORE STRENGTHS

- End-to-end full-stack web & API development
- Database modeling, complex queries & optimization
- Rigorous testing across all SDLC phases
- Industrial IoT & hardware protocol integration
- Fast adoption of new tools & attention to detail
- Strong analytical problem solving & team collaboration

WORK EXPERIENCE

Rudy Technology — 2022 – Present

Full-Stack & Backend Developer (Bangkok, Thailand)

- Delivered end-to-end full-stack web solutions and backend services, connecting modern reactive frontends with high-throughput RESTful APIs and microservices.
- Engineered core RESTful APIs serving thousands of concurrent web and mobile clients with sub-100ms response times.
- Architected and optimized relational database schemas in **MySQL & Oracle DB**, written complex queries, stored procedures, and resolved query bottlenecks with indexing under MVC patterns.
- Developed custom backend integration bridges and responsive web interfaces connecting enterprise mobile CRM low-code applications with central microservices.
- Owned and maintained **GitLab CI/CD pipelines**, automating test execution, linting, Docker packaging, and zero-downtime server deployments.
- Authored OpenAPI/Swagger specifications for cross-functional API contracts, managed token authentication lifecycle, and maintained **99.9% API uptime**.

Freelance Industrial Automation & HMI — 2023 – 2024

Industrial Automation & Systems Engineer (Independent / Part-Time Project)

- Architected and developed **VBX_APUST** Industrial Automation HMI software controlling automated adhesive dispensing production lines.
- Integrated multi-protocol hardware: communicated with industrial PLC via **Modbus TCP** (16 DI / 16 DO), Keyence barcode scanners (TCP Socket), and Cognex In-Sight 2800 Machine Vision (HTTP REST/TCP).
- Engineered an end-to-end deterministic **Finite State Machine** workflow enforcing safety interlocks and robotic motion coordination.

So-at Solution — 2021

Software Engineer Intern (Co-operative Training)

- Diagnosed and resolved critical web service incidents, identifying root causes across database locks and network timeouts to restore operational integrity.
- Implemented SQL stored procedures, backend MVC controllers, and created automated **Postman test suites** to validate API contracts between staging and production environments.
- Collaborated with UX designers to translate data schemas into intuitive responsive corporate web interfaces.

SCG (Siam Cement Group), Bang Sue — 2020

Engineering Intern (Student Training)

- Audited enterprise database architectures to optimize query execution plans and retrieval throughput.
- Developed industrial IoT telemetry solutions integrating sensor hardware, microcontrollers, and backend data logging.

FEATURED PROJECTS

3rd-Party Mobile API Integration Gateway — 2023
High-throughput API proxy layer orchestrating third-party vendor APIs with caching, token rotation, and rate-limiting.
Node.js • RESTful API • Microservices • Security

Unified Core RESTful API Platform — 2022
Core backend service endpoints with robust auth, transaction handling, schema validation, and Postman test automation.
Node.js • Laravel • MySQL • Postman • CI/CD

AI Visual Similarity & NLP Pipelines — 2020–2023
Computer vision cosine similarity verification (OpenCV) and NLP intent classification pipelines in Python.
Python • OpenCV • NLP • Text Analytics

Web CRM Platform (React & Redux) — 2021–2022
Interactive business CRM application built with React, Redux, React Hook Form, and backend SQL database integration.
React • Redux • TypeScript • CSS • SQL API

EDUCATION HISTORY

SRIPATUM UNIVERSITY — Graduated: 2021

Bachelor of Computer Engineering (B.Eng) • GPA: 3.27

- **President**, Computer and Technology Organization • **Vice President**, Student Council (Samo Sripatum)
- **National Award:** Distinguished IT Student Award for Ethics & Morality (InCIT / nCIT 2020)